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## WatAware – Intelligent Hydration Companion

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This is an excellent report, scoring 49.5 / 50

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# **1. High-Level Description of Project**

## **1.1 Motivation**

Water is essential to our lives and well-being, but less than half of the population actually drink the recommended amount on a daily basis [1] [2] [3]. Research shows that long-term exposure to dehydration is linked to many medical conditions such as fatigue, joint pain, weight gain, headaches, ulcers, high blood pressure, and kidney diseases [2]. While the consequences are severe for long term dehydration, the general public may not regard consistent hydration as a top priority. Furthermore, many may not even be aware of the fact that they are dehydrated [3] [4]. With WatAware, we hope to enable those who do not drink enough water to meet their daily recommended water intake and avoid the consequences of both acute and chronic dehydration.

## **1.2 Project Objective**

Given that a large percentage of the general public do not drink enough water, the objective of WatAware is to design an intelligent water bottle that can monitor daily water intake and help provide feedback to encourage people to drink more water. The user can set a goal for daily water intake, and the bottle will remind the individual if he or she is not drinking enough water.

To achieve this, the WatAware system provides the user with a method of inputting their daily water intake goal directly into the water bottle. Once the goal has been set, the system will then begin to track the water level and periodically notify the user to rehydrate themselves until their daily goal is met. The user is also given the option to export their water intake history into a smartphone application to be presented in a graphical view to better understand their hydration habits.

There are existing solutions that work towards solving the same problem. However, these alternatives rely heavily on the use of a smartphone to synchronize information between a software application and a water bottle. In contrast, WatAware is a standalone system that is fully functional without the need for any intervention from external devices. Since WatAware is aimed to replace traditional water bottles, it easily integrates into the lifestyle of individuals who understand the importance of hydration but are unaware of the amount of water they intake on a daily basis.

## 1.3 Block Diagram

A high level overview of the WatAware system is shown in Figure 1-1 below.

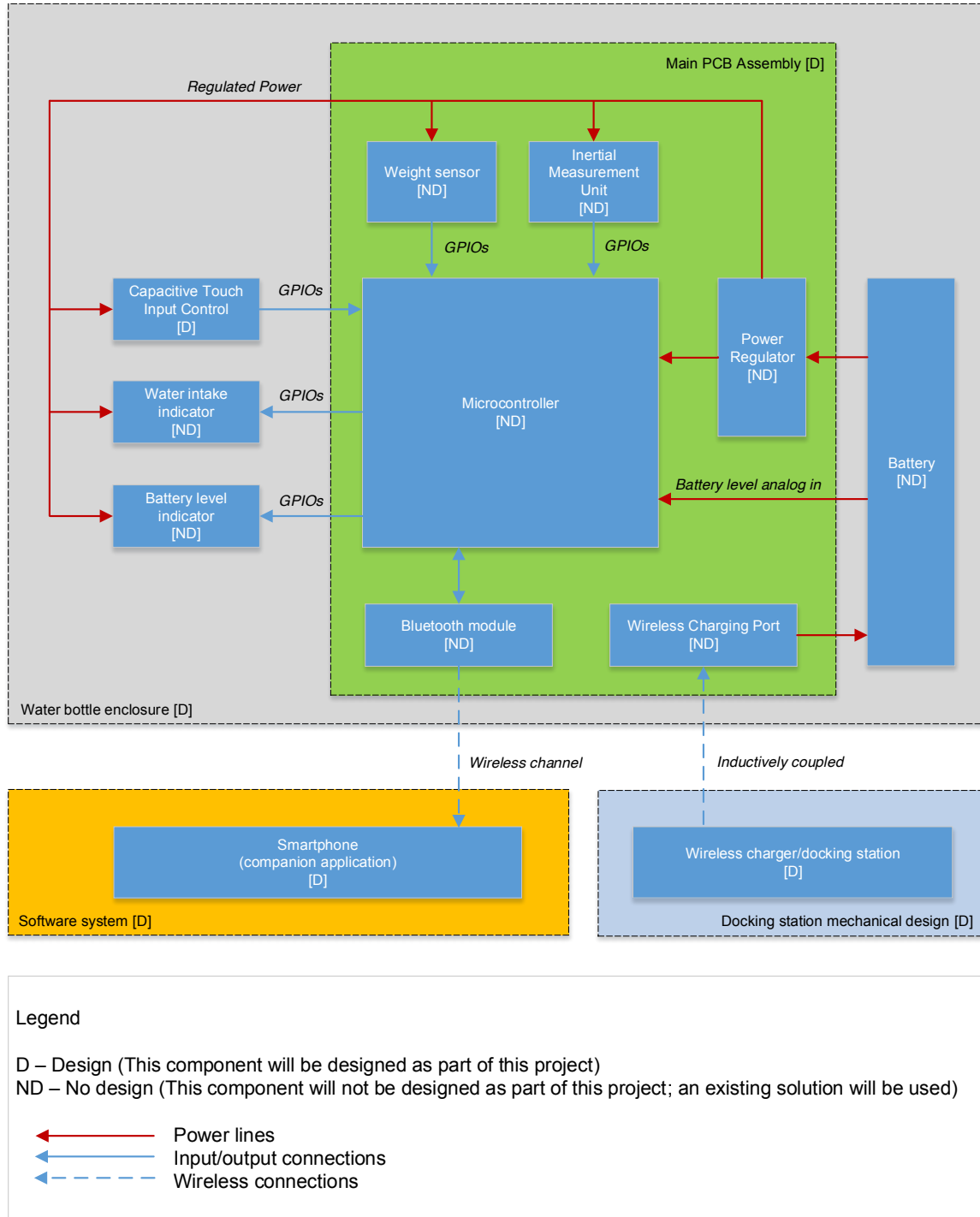


Figure 1-1 High level block diagram

The different components from the block diagram in Figure 1-1 are outlined in Table 1-1 below.

Table 1-1 Description of block diagram components

| <b>Component</b>                 | <b>To Be Designed</b> | <b>Description</b>                                                                                                                                                                                                            |
|----------------------------------|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Capacitive Touch Input Control   | Yes                   | Allows user to set daily water intake goal by sliding the capacitive touch control.                                                                                                                                           |
| Weight sensor                    | No                    | Measures the amount of water that is present in the water bottle. Uses a readily available sensor on the market.                                                                                                              |
| Inertial Measurement Unit        | No                    | Measures the position and movement of the system to determine the state of the system. For example, if the bottle is resting, lifted, tilted or flipped.                                                                      |
| Water intake indicator           | No                    | Shows the amount of water the user has drank relative to his or her daily water intake goal. Uses a readily available display/indicator on the market.                                                                        |
| Battery level indicator          | No                    | Displays the battery level to the user so they know when to charge the system. Uses a readily available display/indicator on the market.                                                                                      |
| Microcontroller                  | No                    | Responsible for handling input data from sensors, outputting values to displays, and communicating with the Bluetooth module to interface with the user's smartphone. Uses a readily available microcontroller on the market. |
| Bluetooth module                 | No                    | Interfaces with the microcontroller to send out information wirelessly. Uses a readily available Bluetooth module on the market.                                                                                              |
| Battery                          | No                    | Provides power for all electronic components within the water bottle. Uses a readily available rechargeable battery solution on the market.                                                                                   |
| Power regulator                  | No                    | Ensures that all electronic components receive a constant voltage source from the battery. Uses a readily available power regulator on the market.                                                                            |
| Wireless charging port           | Yes                   | Inductively couples to the wireless charger, allowing the battery to be charged. This wireless charging system requires design.                                                                                               |
| Wireless charger/docking station | Yes                   | Docking station will begin wirelessly charging the battery in the water bottle once a bottle is placed on top of it. The mechanical and electrical components of this docking station require design.                         |
| Bottle Enclosure                 | Yes                   | The physical assembly of the water bottle that holds the electrical components as well as water. Requires mechanical design of the bottle.                                                                                    |
| Smartphone companion application | Yes                   | Smartphone application communicates with the water bottle via the Bluetooth module. Water intake history is extracted from the water bottle and presented in graphs within the application.                                   |

## 2. Project Specifications

### 2.1 Functional Specifications

The functional specifications that must be satisfied in order to implement the sub-components identified in Figure 1-1 are presented below in Table 2-1.

Table 2-1 Functional Specifications

| <b>Specification</b>                       | <b>Classification</b> | <b>Description</b>                                                                                                                                                                                                                             |
|--------------------------------------------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Volume measurement accuracy                | Essential             | The volume sensor is able to determine the volume of water present in the bottle from 0 to 750mL within 5% accuracy.                                                                                                                           |
| Capacitive Touch Input                     | Essential             | The capacitive touch sensors should be responsive to touch motions such as tapping, sliding and turning for user input.<br>The touch motions on the capacitive touch sensors must be registered within 1 second.                               |
| Inertial Measurement Unit                  | Essential             | The inertial measurement unit must be able to detect motion within 10 degrees of accuracy.                                                                                                                                                     |
| Water intake tracking                      | Essential             | Tracks the history of the water intake through volume differentials. Water intake is only recorded when the bottle cap is fully attached.                                                                                                      |
| Water intake display                       | Essential             | The amount of water the user consumes relative to his or her daily goal is displayed to the nearest 10% rounded down.<br>The display must turn on if any motion is detected. It remains on for 5 seconds before turning off to conserve power. |
| Freshness indicator                        | Non-essential         | The two-digit display shows the amount of time the water has been inside the bottle rounded down to the nearest hour, up to a maximum of 99 hours.                                                                                             |
| Communication                              | Essential             | The bottle must be able to wirelessly communicate with the user's smartphone to send metrics about daily water intake. If this feature is enabled by the user, the metrics are transmitted on an hourly basis.                                 |
| Smartphone application – water intake data | Essential             | The smartphone application must be able to receive and store user water intake history from the bottle and display the data in graphs.                                                                                                         |

## 2.2 Non-functional Specifications

The non-functional specifications that are necessary to complement the functional specifications as discussed in Table 2-1 are shown below in Table 2-2.

Table 2-2 Non-functional Specifications

| Specification           | Classification | Description                                                                                                                                 |
|-------------------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Volume of bottle        | Essential      | The bottle must hold up to 750 mL of drinking water.                                                                                        |
| Size of bottle          | Non-essential  | The dimensions of the bottle will be at most 85mm × 85mm × 280mm.                                                                           |
| Size of dock station    | Non-essential  | The dimensions of the dock station must be at least 95mm × 95mm × 50mm to allow the base of the water bottle to rest on the dock.           |
| Weight of bottle        | Non-essential  | The weight of the bottle must be less than 15 oz.                                                                                           |
| Water resistant         | Essential      | The bottle must be water resistant to allow for washing and can be submerged under water up to a depth of 1 meter.                          |
| Battery capacity        | Essential      | The battery must provide enough power for at least 72 hours of usage regardless of how frequently the bottle is used.                       |
| Battery level indicator | Essential      | The charge remaining in the battery must be displayed with at most 10% error.                                                               |
| Power regulator         | Essential      | The power regular must be able to supply a constant DC voltage that is within 10% of the target value.                                      |
| Wireless charging       | Essential      | The bottle will inductively charge when it is placed on top of the charging dock.                                                           |
| Water Safety            | Essential      | All materials used for the chassis of the water bottle and sensors that come into contact with the water must not contain any known toxins. |

### 3. Risk Assessment

Through the identification of the functional and non-functional specifications for this project, there are multiple associated risks that are summarized below in Table 3-1.

Table 3-1 Risk Assessment and Remediation Strategies

| Potential Risk                | Probability | Impact | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------------------|-------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Obtaining Hardware Components | Medium      | High   | <p><b>Situation</b> – Since the project is heavily hardware focused, many different types of microcontrollers, sensors, regulators, and other components will be prototyped before arriving at the final design. Although one could review the data sheets of each component extensively, many components may prove to be incompatible and the replacement of such components will be required. Depending on the uniqueness and quality of the particular component, there is a risk that a replacement may or may not exist and/or potential shipping delays could gate the implementation of essential features.</p> |
|                               |             |        | <p><b>Remediation</b> – To reduce the likelihood of this scenario, a significant amount of time will be placed during the prototype stage to find multiple suppliers and potential equivalent replacements for each component in our bill of materials. Each component will be verified individually to ensure that the specifications are met prior to integration into the overall system. This will expose any incompatible issues early and allow corrective action to be taken during the early stages of design.</p>                                                                                             |
| Insufficient Time             | Medium      | High   | <p><b>Situation</b> – Given the scope of the project, there are many technical aspects that are predicted to take an extensive amount of time during the design process. All members within the team possess heavy academic course load and are involved with numerous extracurricular activities; as a result, potential delays or missed deadlines could be a product of poor time management.</p>                                                                                                                                                                                                                   |
|                               |             |        | <p><b>Remediation</b> – In order to mitigate this potential risk, the project will be broken down into smaller sub-components, as this will allow us to better analyze the progress of the project. In addition, this will enable members in the group to individually work on different sub-components in parallel. Mandatory number of working hours will be required from each member and the attendance of weekly meetings to discuss progress and gating issues.</p>                                                                                                                                              |



| Potential Risk  | Probability | Impact | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------|-------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Technical Skill | Medium      | Medium | <p><b>Situation</b> – Within the many sub-components of this project, there are many areas which involve the use of new technology that members in the group have had limited to zero exposure and experience in. Areas of which include: The theory behind wireless inductive charging and how to implement this in practice. Wireless network technologies to enable the communication between the water bottle and the smartphone companion application. Physical design of the water bottle and dock station using CAD tools.</p>                                                |
|                 |             |        | <p><b>Remediation</b> – To minimize future roadblocks, the member of which that has the most experience on a certain sub component will drive early design efforts to ensure that reasonable progress can be made. Individual members will also be required to ramp up on all aspects of the project and conduct self-studies to familiarize themselves with areas of which no member in the group is knowledgeable in. The goal of such distribution is to have experts in the team for each sub component rather than having all members focus on one sub component at a time.</p> |

## 4. References

- [1] S. Tucker, "Two-Thirds of Americans Don't Drink Enough," January 2011. [Online]. Available: <http://www.bc.edu/offices/dining/nutrition/topics/drinkenough.html>.
  
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- [4] M. J. McKinley and A. K. Johnson, "The Physiological Regulation of Thirst Intake," *Physiology*, vol. 19, pp. 1-6, 2004.